

VINAYAK GUPTA

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🐙 github.com/vinayakgupta29 | 🔗 linkedin.com/in/vinayak-gupta-70a808202 | 🌐 Portfolio

Professional Summary

Software Engineer with 3 years of experience specializing in robust backend architectures, distributed systems, and cross-platform web/mobile systems using Go, Node.js, and PostgreSQL. Proven track record designing low-latency cryptographic microservices, optimizing concurrent data layers, writing custom multi-platform tooling, and implementing scalable workflows for infrastructure and healthcare platforms supporting over 40k users. Expert in systems design, API engineering, and hardware-software orchestration.

Technical Skills

Languages : Go, TypeScript, JavaScript, Python, SQL, Java, C++, Rust, OCaml
Backend & Systems : Node.js (Express/TS), Django, Echo (Go), REST APIs, Systems Design, Microservices, Cron/Automation Pipelines
Frontend & Mobile : Flutter, Tauri (.deb), React, EJS, HTML5/CSS3, Multilingual Status-Bar Plugins
Databases & Cloud : PostgreSQL (High-Concurrency Optimization), MongoDB, SQLite, AWS (EC2/S3), Docker, Git, Linux (Arch/Nvim customization)

Experience

Techasoft Pvt. Ltd. Bangalore, India
Software Development Intern (Systems & Integration) Mar 2026 – Present

- Engineered high-performance hardware-software control loops using Python (Modbus TCP) and Django, orchestrating low-latency 24V PLC signaling within an 8ms execution window to drive industrial conveyor sorting layers.
- Executed production database migrations from SQLite to PostgreSQL, architecting optimized data schemas to eliminate write-concurrency bottlenecks and stabilize real-time floor-metric logging workflows.

Medoc Health IT Remote
Engineering Product Lead & Project Management Intern Jul 2024 – Jul 2025

- Built a low-latency local Key Management System (KMS) in Go via the Echo framework, implementing PBKDF2 master key derivations to yield sub-2ms cryptographic key retrieval times.
- Navigated legacy government specifications to integrate ABHA v4 health data APIs, mapping complex tokenized JSON schemas for an encrypted data environment serving 40,000+ patients.
- Architected the core scheduling architecture of the MedocEUA application, implementing a sparse-allocation bitwise slot priority algorithm ($2^n - 1$) to manage high-volume emergency triage queuing.
- Authored cross-platform mobile modules in Flutter, integrating background hardware data synchronization protocols via the Android Health Connect API and secure decentralized user consent managers.

Appable Technologies Remote
Freelance Full-Stack Software Developer Jul 2023 – Jun 2024

- Re-architected multiple client applications away from high-cost Firebase structures into custom Node.js and PostgreSQL backends, lowering infrastructure spend by hosting custom web dashboards.
- Designed a recursive organizational hierarchy engine in PostgreSQL to compute nested supervisor permissions and tiered multi-level manager approval lines for an enterprise HRMS.
- Built an offline-first geofenced attendance logging system using camera biometric validation and real-time shift-delta analytics, utilizing automated cron pipelines to process bulk system actions.
- Developed an isolated sports booking platform supporting custom Google Maps markers, dynamic hourly pricing grids, and asynchronous SMS invitation fallbacks for unregistered users.

Medoc Health IT Remote
Software Development Intern Aug 2022 – Jul 2023

- Maintained core healthcare application features using Flutter, Node.js, and SQL, improving deployment efficiency by provisioning containerized environments with custom Docker compose files.

Projects

Zundrath Key Management System | *Go, Cryptography, Echo, PBKDF2* 🐙 Code

- Developed a local cryptographic microservice serving concurrent clients with sub-2ms retrieval latency via mutex-protected on-demand disk indexing.
- Implemented a fault-tolerant, two-phase key rotation pipeline providing a 24-hour safe-migration window to eliminate decryption blackouts during live application schema updates.

Anonymous Live Forum | *Go, html/template, Markdown Parser* 🐙 Code

- Built a stateless, live-hosted textboard using Go's native *html/template* package for server-side rendering, avoiding persistent state overhead.
- Developed a secure tripcode authentication system that hashes user passphrases into unique 16-character identifiers matching strict alphanumeric ranges ($[A-Z][a-z][0-9]$).

Grounded Object Data (GOD) Format | *Go, EBNF Grammar, Reflection API* 🐙 Code

- Designed and engineered a lightweight, schema-less data serialization format utilizing a strict EBNF grammar specification to eliminate JSON payload redundancy and reduce data transmission footprints.
- Implemented a hand-written lexical scanner and recursive-descent parser in Go, introducing native, first-class tabular structures (`header:rows`) that strip repetitive object keys from multi-row payloads.

Education

B.Tech in Computer Science and Engineering 2025
LNCT University, Bhopal